

Projet de débruitage d'images

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M2 IMAGINE



Contexte



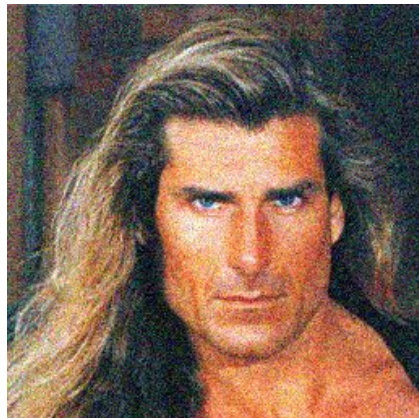
Plan

- État de l'art
- Méthodes proposées
- Résultats
- Démonstration
- Conclusion

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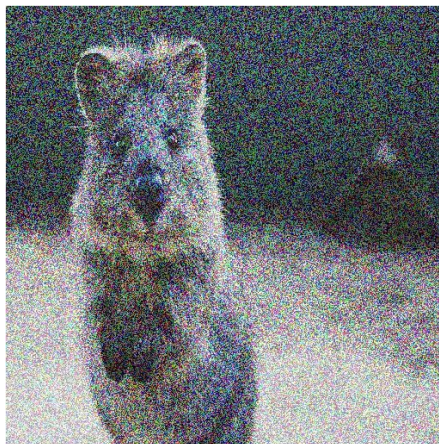
Différents Bruits



Bruit Gaussien



Bruit sel et poivre



Bruit de Poisson

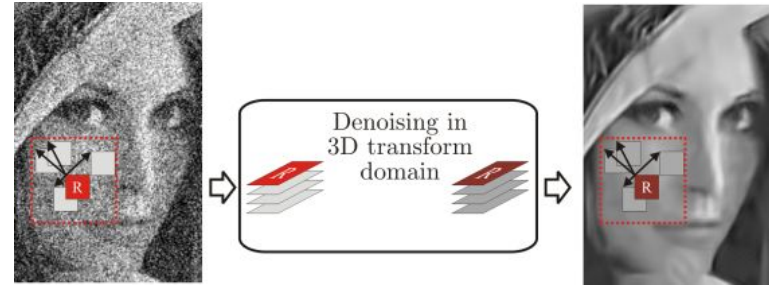


Bruit périodique

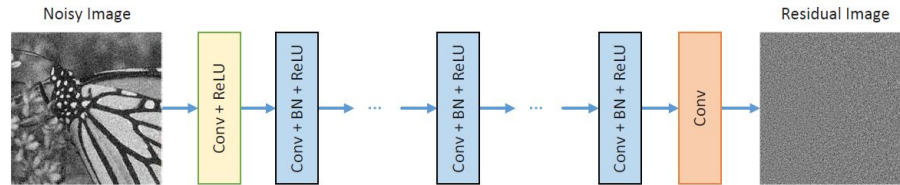
Débruitage



Filtre médian

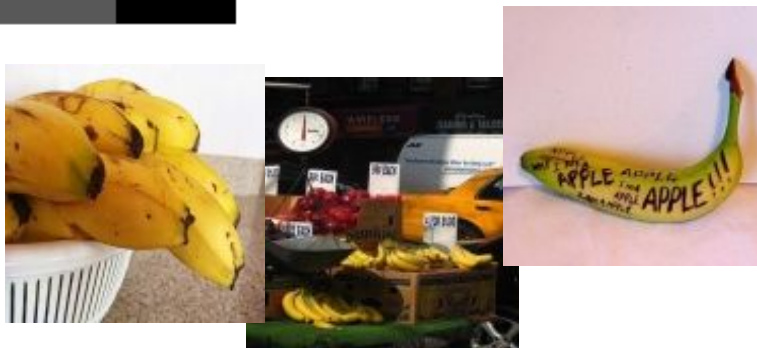


BM3D (Block-Matching and 3D filtering)

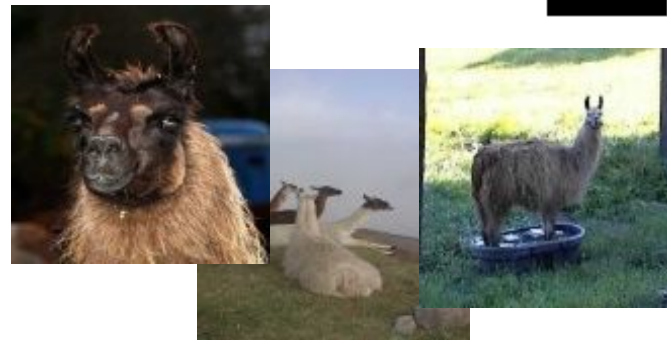


DnCNN

Dataset



bananes



lamas

images.cv



oranges

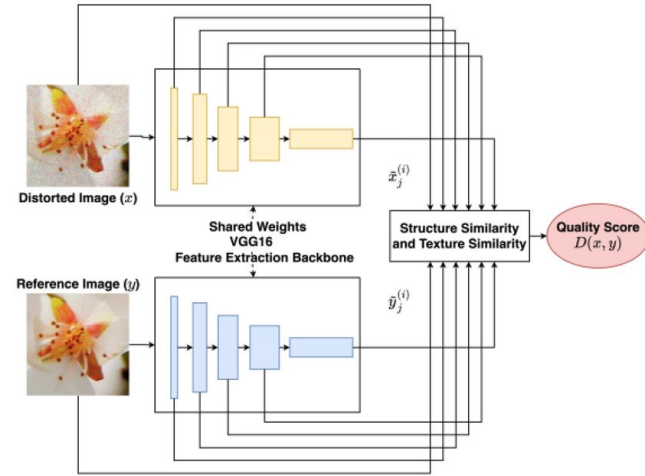
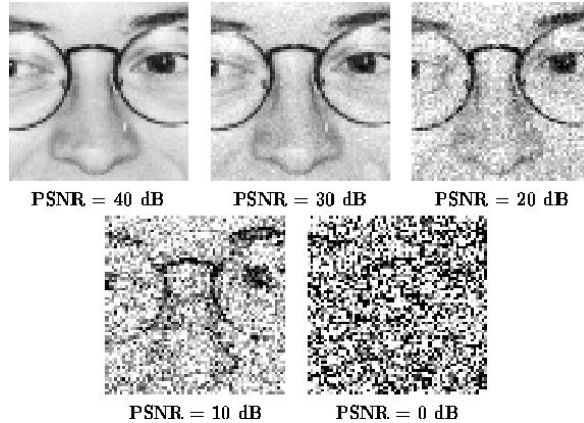


falaises

Métriques



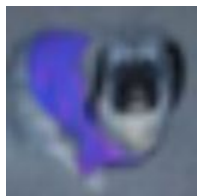
Structural Similarity Index (SSIM)



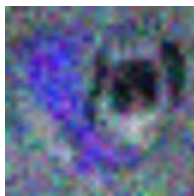
Deep Image Structure and Texture Similarity

VAE

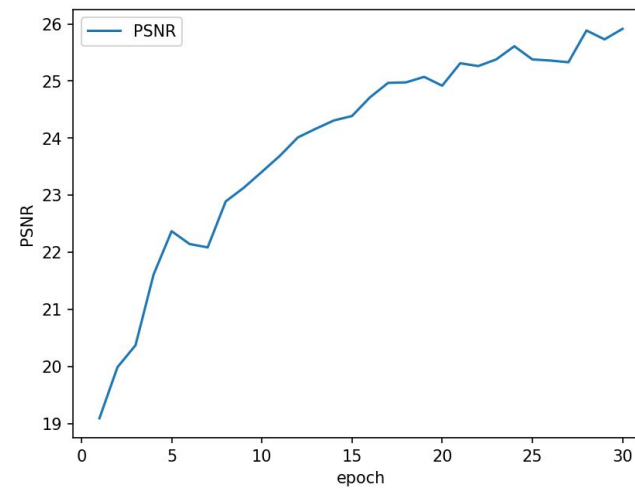
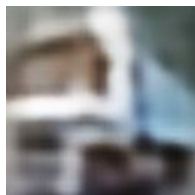
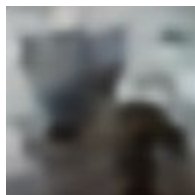
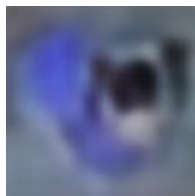
vérité



bruité



prédit



VAE

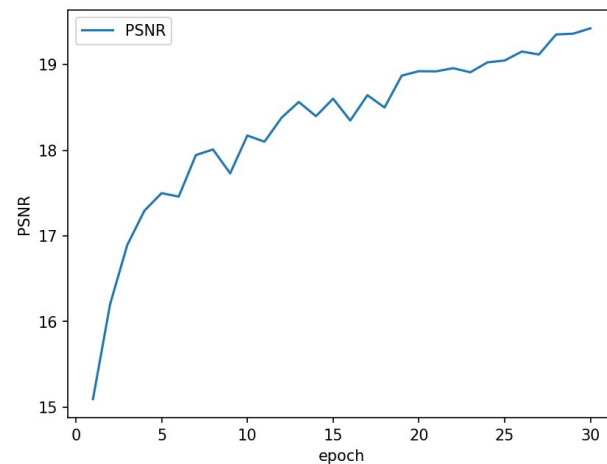
vérité



bruité



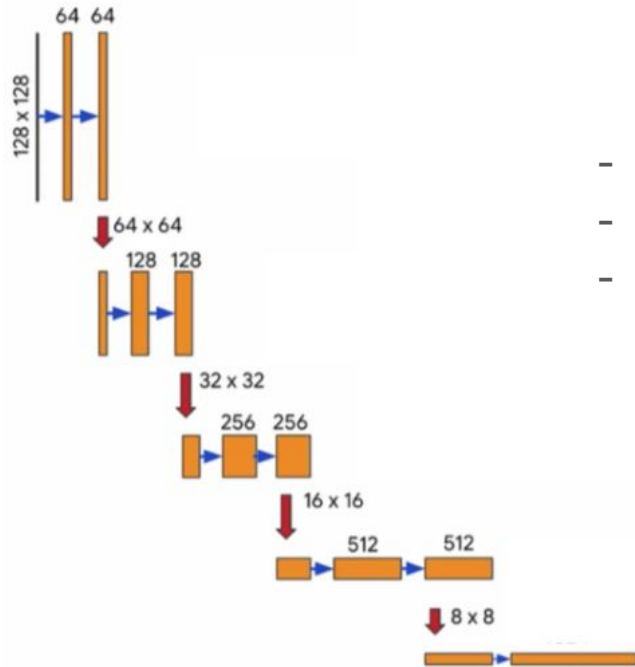
prédit



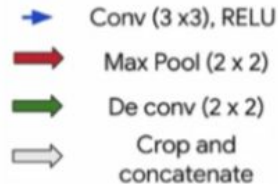
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U-Net Codeur

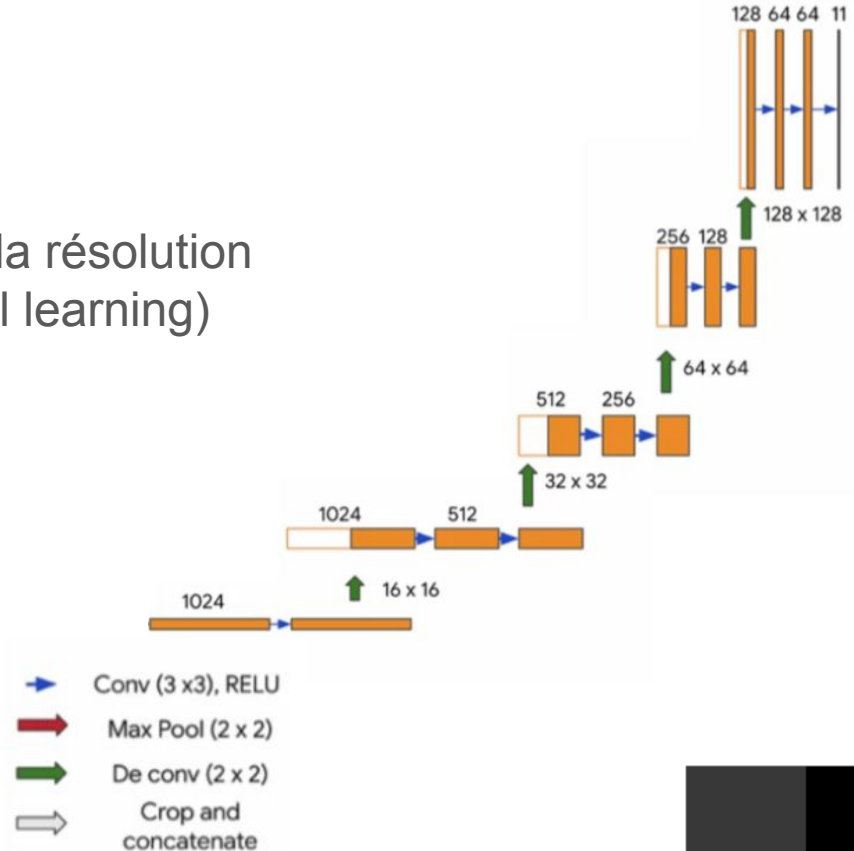


- Extraction des caractéristiques
- Réduction progressive de la résolution
- Produit une représentation compacte

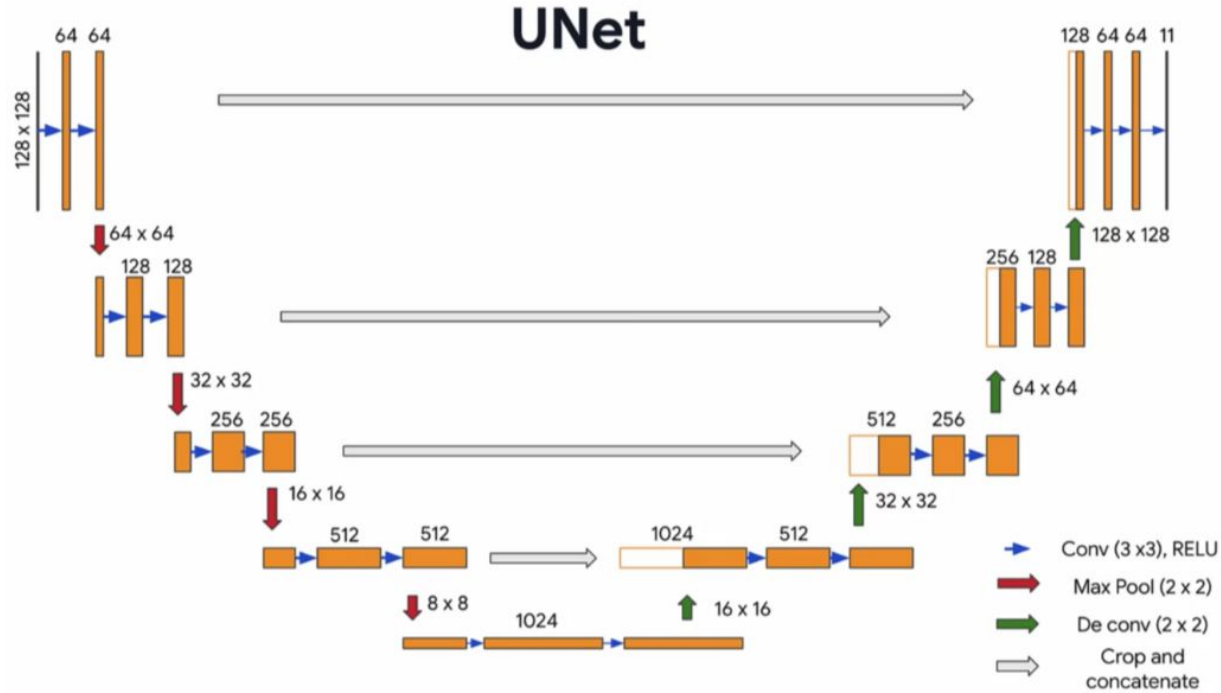


U-Net Décodeur

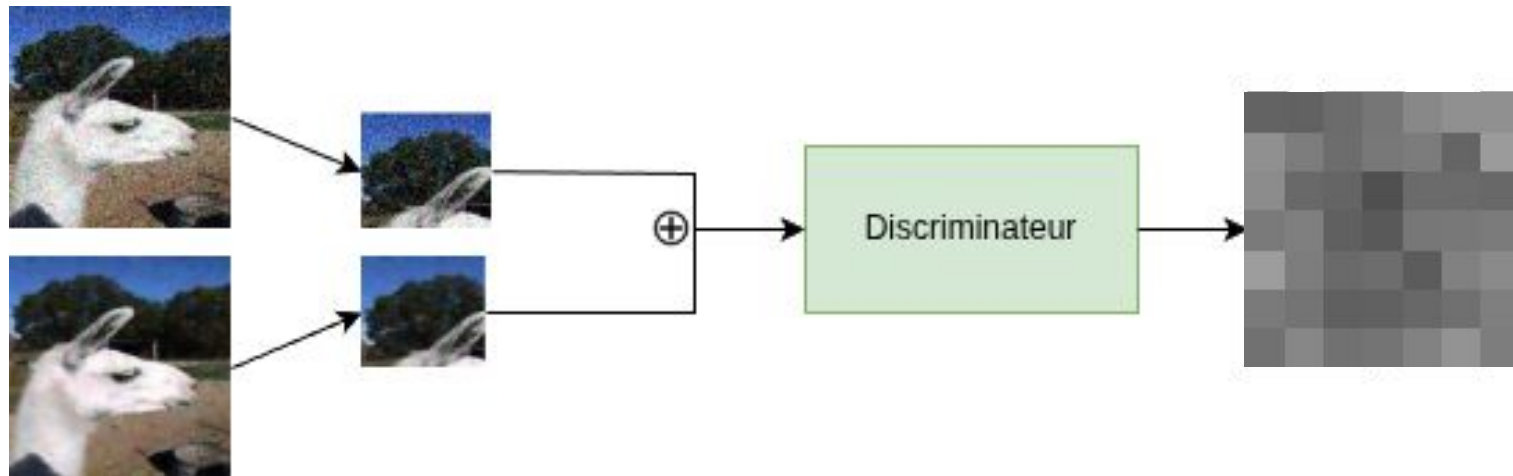
- Reconstruction progressive
- Upsampling pour remonter la résolution
- Prédit la correction (residual learning)



U-Net



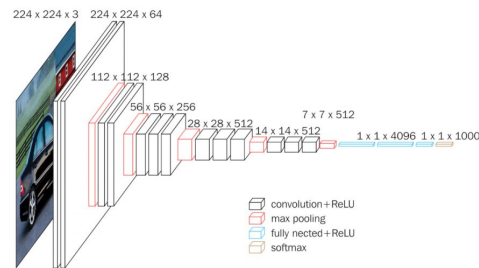
Patch GAN



Loss

$$\sum_{i=1}^n |y_{true} - y_{predicted}|$$

L1



VGG

$$\ell(y) = \max(0, 1 - t \cdot y)$$

Hinge

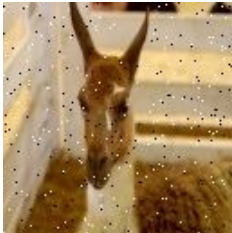
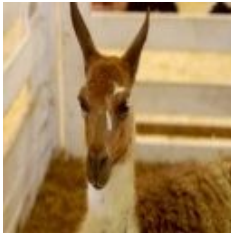
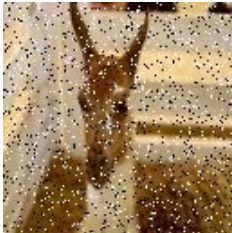
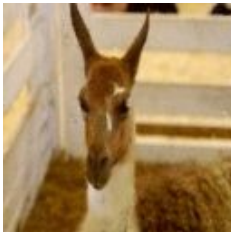
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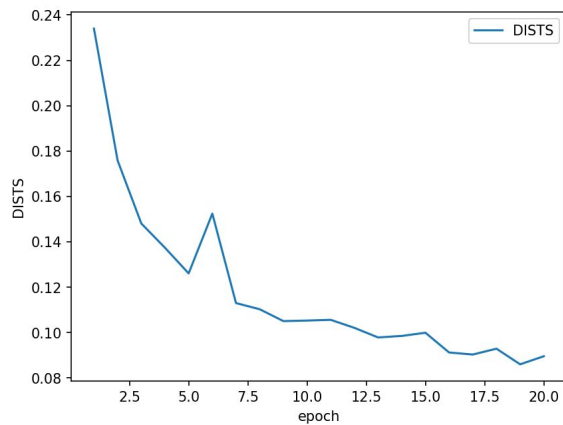
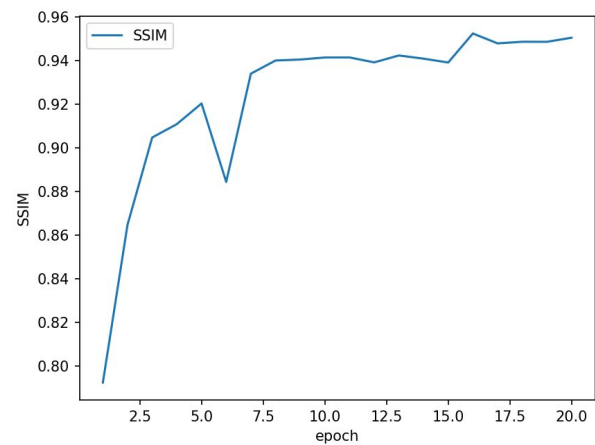
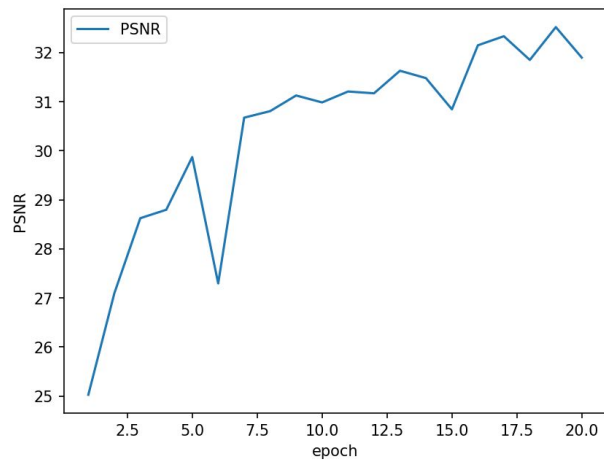
Résultats

- 20-25 epochs
- Batch size 16
- Pré-entraînement d'U-Net : 10 epochs
- Décroissance de L1

Sel et Poivre

vérité	bruité	prédit	PSNR	SSIM	DISTS
			36.86 dB	0.978	0.0468
			34.28 dB	0.959	0.0722
			32.43 dB	0.949	0.0953

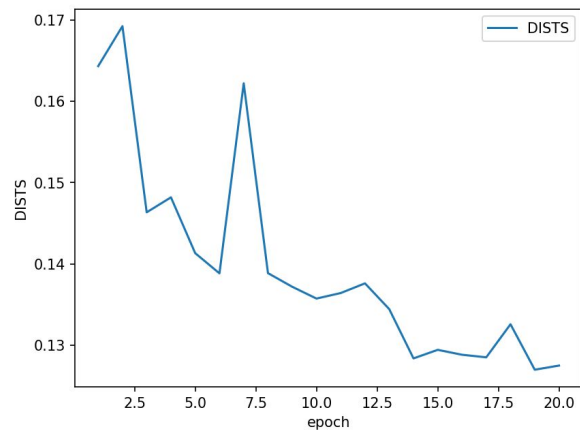
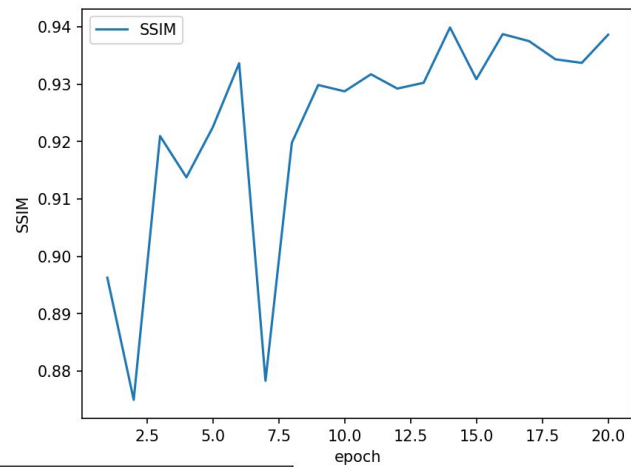
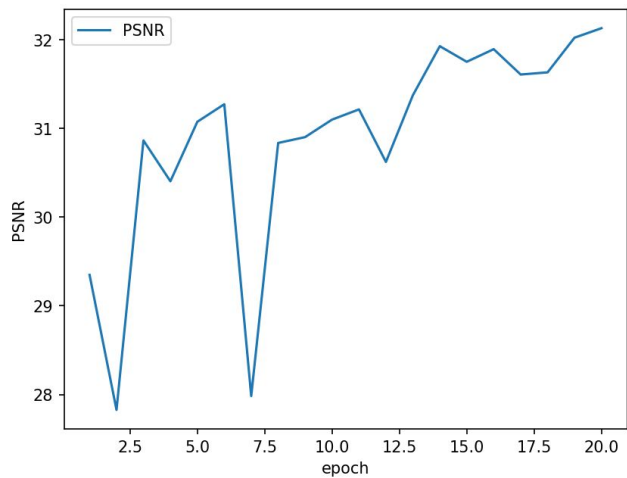
Sel et Poivre



Poisson

vérité	bruité	prédit	PSNR	SSIM	DISTS
			32.55 dB	0.942	0.1177
			31.70 dB	0.937	0.1271
			30.25 dB	0.918	0.1467

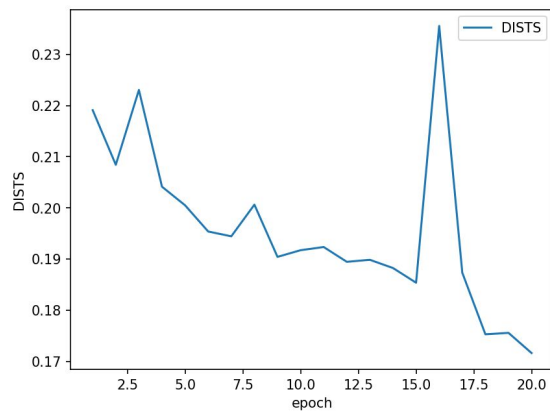
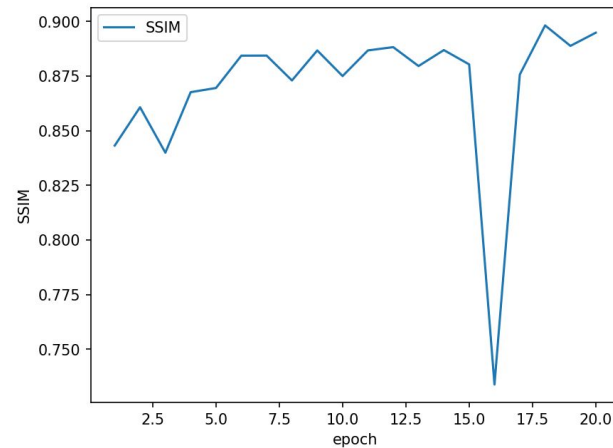
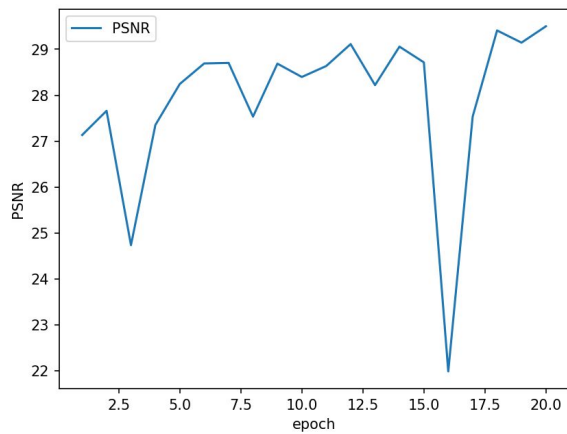
Poisson



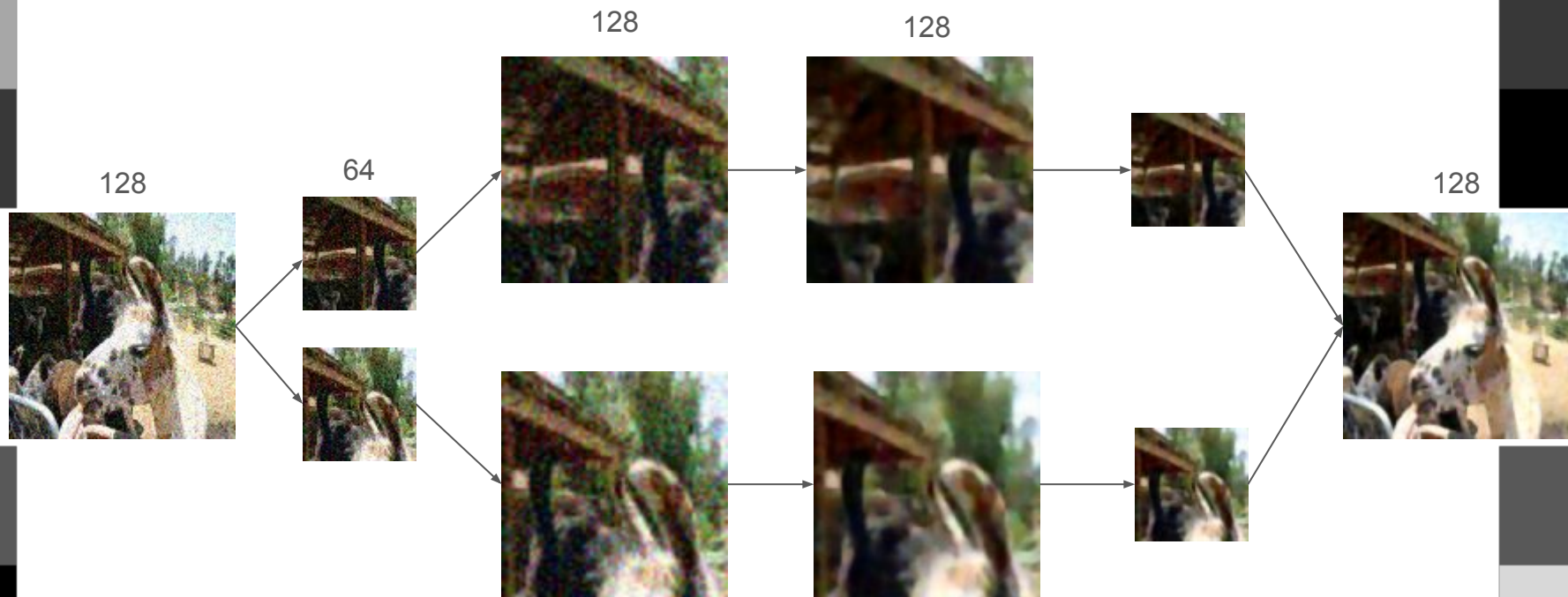
Gaussien

	vérité	bruité	prédit	PSNR	SSIM	DISTS
$\sigma = 5$				36.74 dB	0.973	0.0624
$\sigma = 25$				29.43 dB	0.896	0.1696
$\sigma = 50$				26.54 dB	0.835	0.2125

Gaussien



Gaussian Tiling

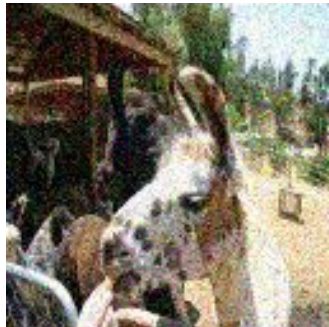


Gaussian Tiling

vérité



bruité



Classique



Tiling



PSNR : 29.66 dB

29.91 dB

SSIM : 0.8609

0.8756

DISTS : 0.1650

0.2209

Gaussian Tiling



Gaussian Tiling

vérité



bruité



Classique



Tiling



PSNR : 30.32 dB

30.32 dB

SSIM : 0.8504

0.8502

DISTS : 0.1370

0.1368

Comparatif

Périodique > Sel et Poivre > Poisson > Gaussien

Bruit périodique
PSNR ++

Bruit aléatoire
PSNR +

Bruit "lumineux"
PSNR -

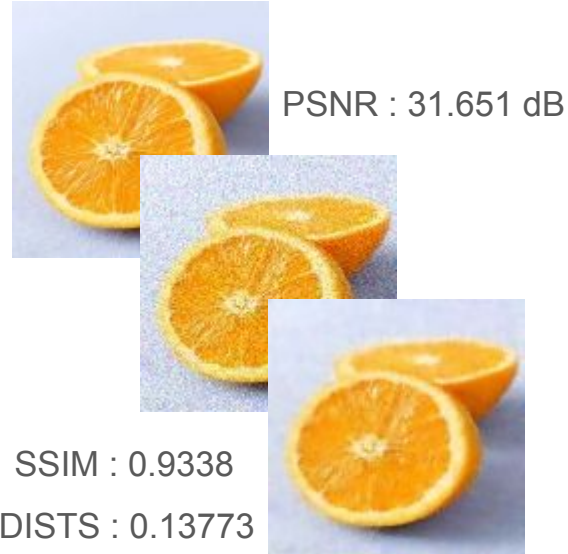
Bruit parasite fort
PSNR --

Mix de bruits

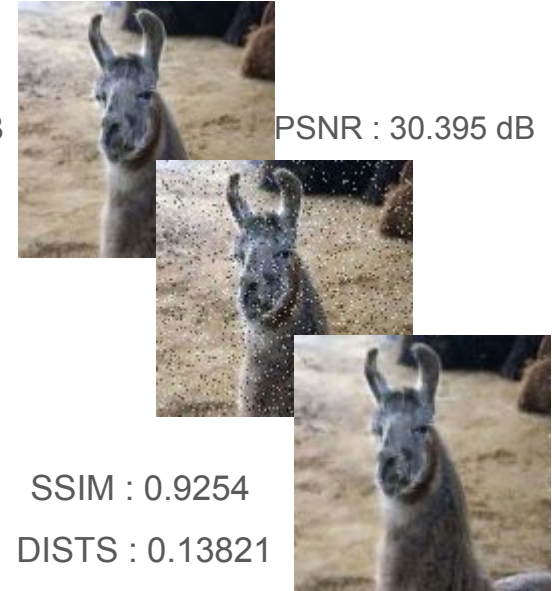
Gaussien + Sel et Poivre



Gaussien + Poisson



Sel et Poivre + Poisson

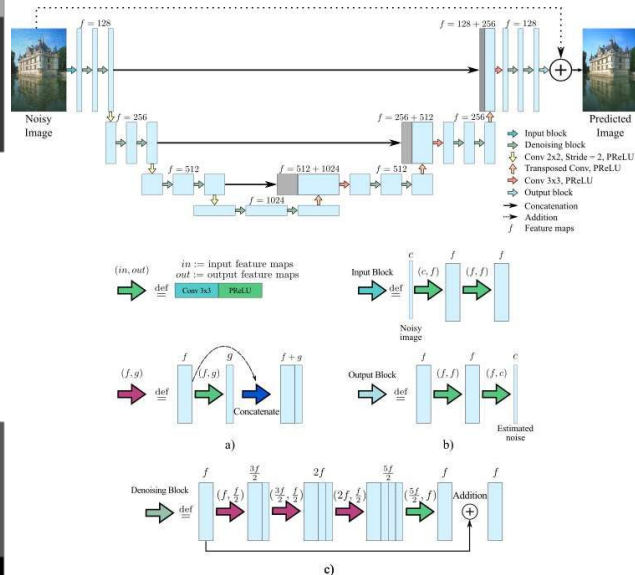


Gaussien : 29.66 dB | Sel et Poivre : 34.28 dB | Poisson : 31.70 dB

Périodique

vérité	bruité	prédit	PSNR	SSIM	DISTS
			41.07 dB	0.989	0.0255
			37.56 dB	0.976	0.0595
			33.07 dB	0.954	0.1061

RDU-Net



vérité



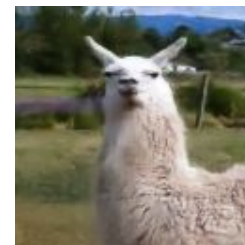
bruité



U-Net



RDU-Net



PSNR : 29.433 dB

29.572 dB

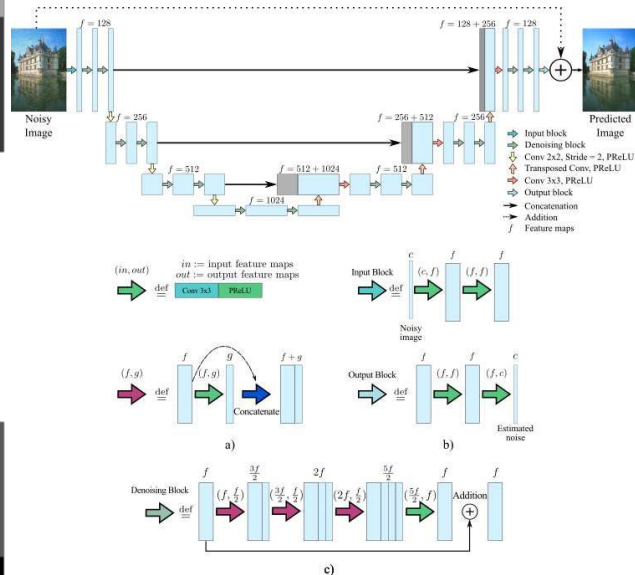
SSIM : 0.896

0.906

DISTS : 0.1696

0.1549

RDU-Net



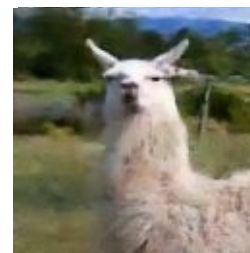
vérité



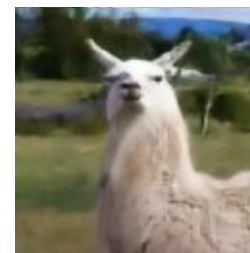
bruité



U-Net



RDU-Net



PSNR : 26.543 dB

26.723 dB

SSIM : 0.835

0.838

DISTS : 0.2126

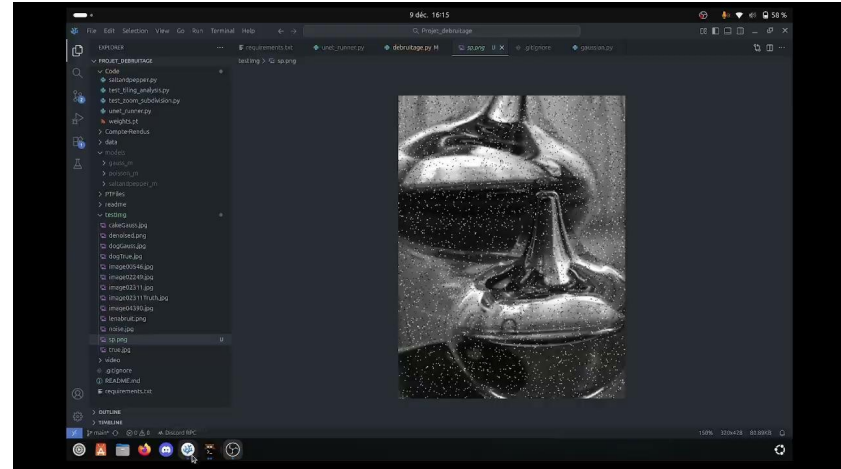
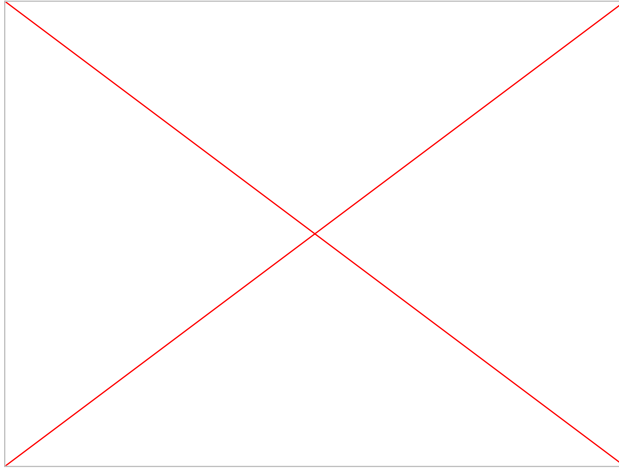
0.2100

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Démo

Débruitage d'une image



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Conclusion

- Interface
- Encore plus d'images !
- Des grandes images !



Merci !

Références

- [Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising](#)
2016 - Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, et Lei Zhang
- [Image Denoising Using a U-net](#)
2019- Paavani Dua
- [A Residual Dense U-Net Neural Network for Image Denoising](#)
2021- Javier Gurrola-Ramos, Oscar Dalmau, et Teresa E. Alarcon
- [GAN-based Noise Model for Denoising Real Images](#)
2020- Linh Duy Tran, Son Minh Nguyen, et Masayuki Arai

Github :

- <https://github.com/SecretMG/UNet-for-Image-Denoising>
- <https://github.com/JavierGurrola/RDUNet>

Bruits :

- [Gaussian Noise](#)
- [Poivre et Sel](#)
- [Poisson](#)
- [CIFAR-10](#) / images.cvy : Dataset d'images